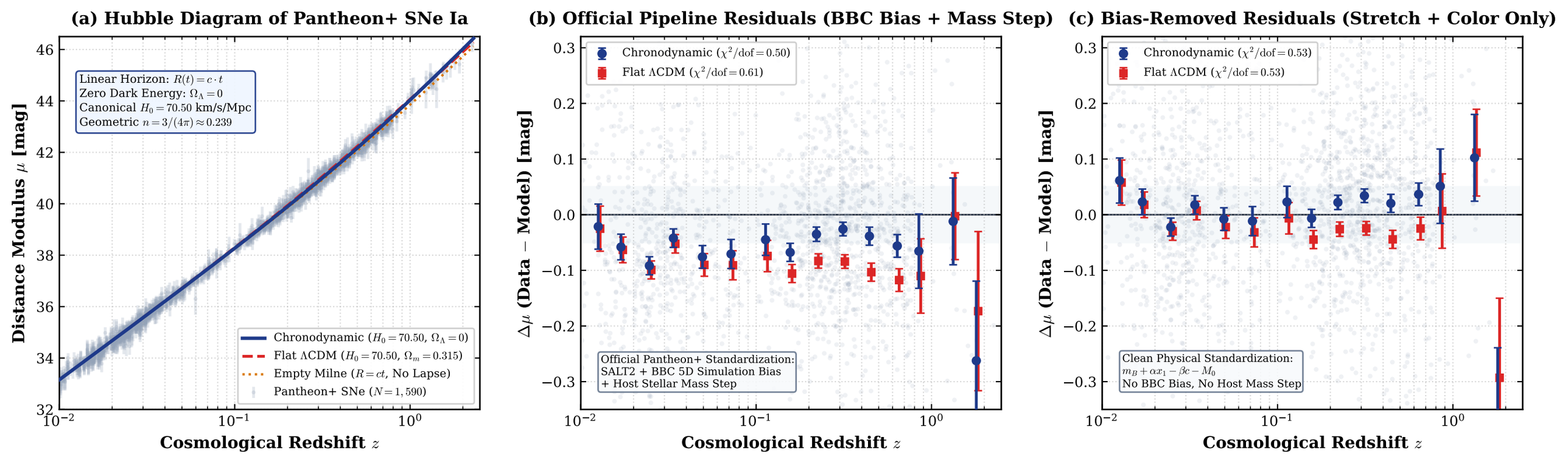

(a) Hubble Diagram of Pantheon+ SNe Ia
Linear Horizon: $R(t) = c \cdot t$
Zero Dark Energy: $\Omega_\Lambda = 0$
Canonical $H_0 = 70.50$ km/s/Mpc
Geometric $n = 3/(4\pi) \approx 0.239$
Chronodynamic ($H_0 = 70.50$, $\Omega_\Lambda = 0$)
Flat ΛCDM ($H_0 = 70.50$, $\Omega_m = 0.315$)
Empty Milne ($R = ct$, No Lapse)
Pantheon+ SNe ($N = 1,590$)
Distance Modulus μ [mag]
Cosmological Redshift z
(b) Official Pipeline Residuals (BBC Bias + Mass Step)
Chronodynamic (χ^2/dof = 0.50)
Flat ΛCDM (χ^2/dof = 0.61)
Official Pantheon+ Standardization:
SALT2 + BBC 5D Simulation Bias
+ Host Stellar Mass Step
$\Delta\mu$ (Data − Model) [mag]
Cosmological Redshift z
(c) Bias-Removed Residuals (Stretch + Color Only)
Chronodynamic (χ^2/dof = 0.53)
Flat ΛCDM (χ^2/dof = 0.53)
Clean Physical Standardization:
$m_B + \alpha x_1 - \beta c - M_0$
No BBC Bias, No Host Mass Step
$\Delta\mu$ (Data − Model) [mag]
Cosmological Redshift z